

Human-Computer Interaction — Lecture 03: The Computer

المحاضرة 3: الحاسوب (The Computer)

Multiple Choice (25)

1. According to the lecture, the devices used in a computer system dictate what?

حسب المحاضرة، الأجهزة المستخدمة في نظام الحاسوب بتحدد إيه؟

- A. The styles of interaction the system supports
- B. The price of the system
- C. The speed of the network
- D. The size of the memory

✓ الإجابة الصحيحة: A) The styles of interaction the system supports

ليه صح: الأجهزة هي اللي بتحدد أنماط التفاعل اللي النظام يدعمها، ولو غيّرنا الأجهزة يتغير نمط التفاعل. السعر والسرعة والذاكرة مش هي اللي المحاضرة بتربطها بنمط التفاعل.

المرجع: Lecture 03 — Slide 4

2. In the era of batch processing, which description is correct?

في عصر المعالجة الدفعية (batch processing)، أي وصف صح؟

- A. Punched card stacks were prepared, followed by a long wait for line printer output
- B. Users received rapid feedback and were in control
- C. Voice commands were used to run programs
- D. Touch screens were the main input device

✓ الإجابة الصحيحة: A) Punched card stacks were prepared, followed by a long wait for line printer output

ليه صح: في ال batch كانوا يجهزوا كروت مثقبة وبعدها انتظار طويل لمخرجات الطابعة. وصف التغذية الراجعة السريعة وتحكم المستخدم بيخص الحوسبة التفاعلية الحالية مش ال batch.

المرجع: Lecture 03 — Slide 7

3. Why is the QWERTY layout described as not optimal for typing?

ليه ترتيب QWERTY مش الأمثل للكتابة؟

- A. It was designed to prevent typewriters from jamming
- B. It places vowels far from the home row on purpose
- C. It was designed for one-handed use
- D. It was created for phone pads

✓ الإجابة الصحيحة: A) It was designed to prevent typewriters from jamming

ليه صح: الترتيب اتصمم أصلاً عشان يمنع تعليق الآلات الكاتبة. باقي الاختيارات مش سبب ذكره الدكتور.

المرجع: Lecture 03 — Slide 12

4. Which keyboard layout places common letters under the dominant fingers and alternates common letter combinations between hands?

أي ترتيب كيورد يحيط الحروف الشائعة تحت الأصابع القوية ويبذل بين اليدين في تركيبات الحروف؟

- A. Dvorak
- B. Alphabetic
- C. QWERTY
- D. Chord

✓ الإجابة الصحيحة: A) Dvorak

ليه صح: ده تعريف Dvorak ويعطي تحسن 10-15% في السرعة. الـ Alphabetic مش أسرع لا للمحترفين ولا للمبتدئين.

المرجع: Lecture 03 — Slide 15

5. Which statement about chord keyboards is correct?

أي عبارة عن chord keyboards صحيحة؟

- A. They have a full set of 100+ keys
- B. Letters are typed as a combination of keypresses on only a few keys
- C. They are the slowest text-entry device
- D. They need no learning time and have no fatigue issues

✓ الإجابة الصحيحة: B) Letters are typed as a combination of keypresses on only a few keys

ليه صح: الـ chord keyboard فيه 4 أو 5 مفاتيح بس، والحرف بيتكتب بمجموعة ضغطات. هو سريع بعد التدريب وحجمه صغير، لكن فيه مقاومة اجتماعية وتعب بعد الاستخدام الطويل.

المرجع: Lecture 03 — Slide 17

6. In T9 predictive text entry, how is the word guessed?

في T9، إزاي بيتخمن الكلمة؟

- A. By pressing each key multiple times
- B. By recognizing the user's handwriting
- C. By measuring pressure on the key
- D. By typing one key per letter and using a dictionary to guess the word

✓ الإجابة الصحيحة: D) By typing one key per letter and using a dictionary to guess the word

ليه صح: في T9 بتضغط مفتاح واحد لكل حرف والنظام بيستخدم قاموس يخمن الكلمة. الضغط المتكرر هو الطريقة العادية للـ phone pad قبل T9.

المرجع: Lecture 03 — Slide 19

7. Speech recognition works best in which situation?

التعرف على الكلام يشتغل أحسن في أنهي حالة؟

- A. Many speakers with a large vocabulary in a noisy room
- B. A single trained user with a limited vocabulary
- C. Any user with imprecise pronunciation
- D. Several users speaking at once

✓ **الإجابة الصحيحة:** B) A single trained user with a limited vocabulary

ليه صح: بينج أكثر مع مستخدم واحد مدَّرب ومفردات محدودة. الضوضاء وتعدد المتحدثين والمفردات الكبيرة كلها مشاكل.

المرجع: Lecture 03 — Slide 21

8. Why is the mouse called an indirect manipulation device?

ليه الماوس اسمه جهاز تحكم غير مباشر؟

- A. Mouse movement is in the (x, z) plane while the cursor moves in the (x, y) plane
- B. It needs a cable to connect
- C. It cannot detect movement
- D. It obscures the screen

✓ **الإجابة الصحيحة:** A) Mouse movement is in the (x, z) plane while the cursor moves in the (x, y) plane

ليه صح: حركة الماوس في المستوى (x, z) بينما المؤشر على الشاشة في (x, y). الماوس مش بيحجب الشاشة، وده من مميزاته.

المرجع: Lecture 03 — Slide 25

9. What is an advantage of optical mice over mechanical mice?

إيه ميزة الماوس الضوئي مقارنة بالميكانيكي؟

- A. They require a ball on the underside
- B. They only work on a special grid pad
- C. They are less susceptible to dust and dirt
- D. They detect absolute position

✓ **الإجابة الصحيحة:** C) They are less susceptible to dust and dirt

ليه صح: الضوئي أقل تأثراً بالتراب. الكرة موجودة في الميكانيكي، والماوس بيكتشف الحركة النسبية بس مش المطلقة.

المرجع: Lecture 03 — Slide 26

10. Which is a disadvantage of touch-sensitive screens?

أي واحدة من عيوب شاشات اللمس؟

- A. They require a specialised pointer
- B. They are imprecise for small regions and the finger can mark the screen
- C. They are slow for menu selection
- D. They cannot be used in hostile environments

✓ **الإجابة الصحيحة:** B) They are imprecise for small regions and the finger can mark the screen

ليه صح: الإصبع مش دقيق ويسيب علامات على الشاشة. باقي الاختيارات غلط لأن الشاشات اللمسية مش محتاجة مؤشر خاص، وسريعة في اختيار القوائم، ومناسبة للبيئات القاسية.

المرجع: Lecture 03 — Slide 31

11. What is the main use of eyegaze today according to the lecture?

إيه الاستخدام الرئيسي لتتبع النظر (eyegaze) حسب المحاضرة؟

- A. Mainly for gaming
- B. Mainly for printing
- C. Mainly for text entry on phones
- D. Mainly for evaluation, with potential for hands-free control

✓ **الإجابة الصحيحة:** D) Mainly for evaluation, with potential for hands-free control

ليه صح: يُستخدم أساساً للتقييم، ولديه إمكانية للتحكم بدون استخدام اليدين. الدقة العالية محتاجة headset.

المرجع: Lecture 03 — Slide 34

12. What does 'colour depth' mean?

إيه معنى عمق اللون (colour depth)؟

- A. The density of pixels per inch
- B. The ratio between width and height
- C. The number of pixels on the screen
- D. How many different colours each pixel can have

✓ **الإجابة الصحيحة:** D) How many different colours each pixel can have

ليه صح: عمق اللون = عدد الألوان المختلفة لكل بكسل (مثلاً 8 بت لكل لون من RGB = ملايين الألوان). كثافة البكسل هي الـ dpi، والنسبة بين العرض والارتفاع هي aspect ratio.

المرجع: Lecture 03 — Slide 39

13. What does anti-aliasing do?

إيه وظيفة ال anti-aliasing ؟

- A. Makes lines thicker
- B. Increases the refresh rate
- C. Softens jagged edges by using shades of the line colour
- D. Removes colours from the screen

✓ **الإجابة الصحيحة:** C) Softens jagged edges by using shades of the line colour

ليه صح: بيخفف حواف ال jaggies باستخدام درجات من لون الخط، ويستخدم مع النصوص كمان.

المرجع: Lecture 03 — Slide 40

14. In a cathode ray tube (CRT), what makes the screen glow?

في شاشة CRT، إيه اللي بيخلي الشاشة تتوهج؟

- A. Liquid crystal polarisation
- B. A beam of electrons hitting a phosphor-coated screen
- C. A laser reflected off a mirror
- D. Reflected ambient light

✓ **الإجابة الصحيحة:** B) A beam of electrons hitting a phosphor-coated screen

ليه صح: شعاع إلكترونات من المدفع الإلكتروني يصطدم بشاشة مغطاة بالفوسفور فتتوهج. الاستقطاب البلوري خاص بـ LCD.

المرجع: Lecture 03 — Slide 41

15. Why do LCDs cause less eye strain according to the lecture?

ليه شاشات LCD بتسبب إجهاد عين أقل حسب المحاضرة؟

- A. They are brighter
- B. Light is reflected, not emitted
- C. They use phosphor
- D. They have higher refresh rates

✓ **الإجابة الصحيحة:** B) Light is reflected, not emitted

ليه صح: الضوء بيتعكس مش بيتبعث (N.B. في الشريحة). الفوسفور خاص بـ CRT.

المرجع: Lecture 03 — Slide 44

16. What causes VR motion sickness according to the lecture?

إليه سبب دوار الحركة في الواقع الافتراضي؟

- A. Too much battery usage
- B. Loud sounds in the headset
- C. Conflicting cues such as time delay between head movement and display, and eye angle vs. focus
- D. Low colour depth

✓ **الإجابة الصحيحة:** C) Conflicting cues such as time delay between head movement and display, and eye angle vs. focus

ليه صح: الأشارات المتضاربة: تأخر الشاشة عن حركة الرأس، وتعارض زاوية العين مع مستوى التركيز. الأسباب الأخرى مش مذكورة.

المرجع: Lecture 03 — Slide 55

17. Which device uses fibre optics to detect finger position?

أي جهاز يستخدم الألياف الضوئية لاكتشاف موضع الأصابع؟

- A. Data glove
- B. 3D mouse
- C. VR helmet
- D. Joystick

✓ **الإجابة الصحيحة:** A) Data glove

ليه صح: الـ data glove. الـ 3D mouse فيه ست درجات حرية، والـ VR helmet بيكتشف حركة الرأس.

المرجع: Lecture 03 — Slide 51

18. Devices that provide touch and force feedback, such as vibration in games, are called what?

الأجهزة اللي بتوفر إحساس اللمس وقوة الارتداد زي الاهتزاز في الألعاب اسمها إيه؟

- A. Optical devices
- B. Situated devices
- C. Haptic devices
- D. Raster devices

✓ **الإجابة الصحيحة:** C) Haptic devices

ليه صح: اسمها haptic devices. ومثال ذلك BMW iDrive بيحس المستخدم بـ bumps لكل عنصر في القائمة.

المرجع: Lecture 03 — Slide 60, 61

19. Which printer uses an inked ribbon and a line of pins, with a typical resolution of 80-120 dpi?

أي طابعة تستخدم شريط حبر وصف دبابيس وبدقة 80-120 dpi؟

- A. Laser printer
- B. Dot-matrix printer
- C. Ink-jet printer
- D. Thermal printer

✓ الإجابة الصحيحة: B) Dot-matrix printer

ليه صح: ال dot-matrix. ال ink-jet بدقة 300 dpi أو أفضل، وال laser حوالي 600 dpi أو أكثر، وال thermal تستخدم ورق حساس للحرارة.

المرجع: Lecture 03 — Slide 66, 67

20. What does OCR do?

إيه وظيفة ال OCR؟

- A. Converts text into a bitmap
- B. Compresses images losslessly
- C. Converts a bitmap back into text
- D. Predicts the next word

✓ الإجابة الصحيحة: C) Converts a bitmap back into text

ليه صح: يحول ال bitmap (صورة ممسوحة) لنص. الماسح الضوئي هو اللي يحول الورق لـ bitmap.

المرجع: Lecture 03 — Slide 73, 75

21. Which is true about RAM compared with disks?

أي عبارة صحيحة عن ال RAM مقارنة بالأقراص؟

- A. RAM is non-volatile and slower
- B. RAM stores data for decades without power
- C. RAM is on silicon chips, has fast access (~100 ns) and is usually volatile
- D. RAM is optical

✓ الإجابة الصحيحة: C) RAM is on silicon chips, has fast access (~100 ns) and is usually volatile

ليه صح: ال RAM على شرائح سيليكون وسريعة (حوالي 100 نانو ثانية) وعادة تفقد البيانات بانقطاع الكهرباء (volatile). الأقراص أبطأ (~10ms).

المرجع: Lecture 03 — Slide 78, 79

22. What problem does virtual memory solve, and what is its drawback?

ال virtual memory يحل أنهي مشكلة، وإيه عيبه؟

- A. Not enough RAM; swapping to disk slows things down
- B. Not enough disk space; it makes the CPU hotter
- C. Screen resolution; it blurs text
- D. Network speed; it loses packets

✓ الإجابة الصحيحة: A) Not enough RAM; swapping to disk slows things down

ليه صح: بيخزن بعض البرامج مؤقتاً على القرص فيبان إن ال RAM أكبر، لكن ال swapping (النسخ من القرص لل RAM) بيبطئ النظام.

المرجع: Lecture 03 — Slide 82

23. Which of these is a lossless compression format?

أي واحد من دول بيستخدم ضغط بدون فقدان (lossless)؟

- A. JPEG
- B. MP3
- C. MPEG lossy audio
- D. GIF

✓ الإجابة الصحيحة: D) GIF

ليه صح: ال GIF و ZIP من أمثلة lossless. ال JPEG و MP3 من أمثلة lossy بيستغلوا حدود الإدراك الحسي.

المرجع: Lecture 03 — Slide 83

24. According to Moore's law as described in the lecture, how often does processor speed double?

حسب قانون مور كما في المحاضرة، كل قد إيه بتتضاعف سرعة المعالج؟

- A. Every 6 months
- B. Every 12 months
- C. Every 5 years
- D. Every 18 months

✓ الإجابة الصحيحة: D) Every 18 months

ليه صح: كل 18 شهر. الذاكرة بتتضاعف كل 12 شهر فمدة ال 12 شهر خاصة بالذاكرة مش المعالج.

المرجع: Lecture 03 — Slide 89

25. Which limitation is described as 'bottleneck in transference of data from disk to memory'?

أي قيد وُصف بأنه 'عنق زجاجة في نقل البيانات من القرص للذاكرة'؟

- A. Computation bound
- B. Graphics bound
- C. Network capacity
- D. Storage channel bound

✓ **الإجابة الصحيحة:** D) Storage channel bound

ليه صح: ده ال Storage channel bound. ال Computation bound يعني الحساب بياخد وقت طويل، وال Graphics bound عن تحديث الشاشة.

المرجع: Lecture 03 — Slide 91

Essay and Calculation Questions (17)

1. List the elements of a computer system that affect interaction, and explain why we need to understand computers in HCI.

اذكر عناصر نظام الحاسوب المؤثرة على التفاعل ولية لازم نفهم الحاسوب في HCI.

Model answer:

Elements: input devices (text entry and pointing), output devices (screens, digital paper), virtual reality (special interaction and display devices), physical interaction (sound, haptic, bio-sensing), paper (print and scan), memory (RAM and permanent media), processing (speed, networks).

Each element affects the interaction. To understand HCI we need to understand what goes in and out (devices, paper, sensors) and what the computer can do (memory, processing, networks). The devices dictate the styles of interaction the system supports.

المرجع: Lecture 03 — Slide 2-4

2. Compare the QWERTY, alphabetic, and Dvorak keyboard layouts, and explain why QWERTY remains dominant.

قارن بين ترتيبات QWERTY و Alphabetic و Dvorak ولية QWERTY لسه مسيطر.

Model answer:

QWERTY: standardised layout, but not optimal for typing because it was designed to prevent typewriters jamming.

Alphabetic: keys in alphabetic order; not faster for trained typists or beginners.

Dvorak: common letters under dominant fingers, biased towards the right hand, common letter combinations alternate between hands; 10-15% improvement in speed and less fatigue.

QWERTY remains dominant because the large social base of QWERTY typists creates market pressure and reluctance to change.

المرجع: Lecture 03 — Slide 12-15

3. Explain chord keyboards and phone-pad text entry (multi-tap and T9).

اشرح chord keyboards والكتابة على لوحة الهاتف (multi-tap و T9).

Model answer:

Chord keyboards: only 4 or 5 keys; letters typed as combinations of keypresses; compact (ideal for portable use), short learning time, fast once trained; but social resistance and fatigue after extended use; a niche market for some wearables.

Phone pad: numeric keys pressed multiple times (e.g. hello = 4433555 [pause] 555666), surprisingly fast. T9 predictive entry: type one key per letter and a dictionary guesses the word (hello = 43556); ambiguities such as 26 produce a menu of 'am' or 'an'.

المرجع: Lecture 03 — Slide 17-19

4. Compare the pointing devices: mouse, touchpad, trackball, joystick, and touch screen. Which are direct and which are indirect?

قارن بين أجهزة الإشارة وحدد المباشر وغير المباشر.

Model answer:

Mouse: handheld, planar movement and buttons; relative movement only; indirect manipulation (mouse moves in the x-z plane, cursor in x-y); does not obscure the screen, accurate and fast; hand-eye coordination problems for novices; mechanical (ball) or optical (LED, less susceptible to dust).

Touchpad: small touch-sensitive tablet stroked to move the pointer, used in laptops; good acceleration settings are important.

Trackball: ball rotated inside a static housing, like an upside-down mouse; indirect, fairly accurate, very fast for gaming.

Joystick: indirect; pressure of stick = velocity of movement; used for games, aircraft controls and 3D navigation.

Touch screen: direct pointing device; fast, no specialised pointer, good for menu selection; but the finger marks the screen, is imprecise, and lifting the arm is tiring. Stylus and light pen are also direct but can obscure the screen.

المرجع: Lecture 03 — Slide 24-32

5. Compare CRT and LCD displays, including how they work and the health considerations.

قارن بين شاشات CRT و LCD من حيث طريقة العمل والاعتبارات الصحية.

Model answer:

CRT: a stream of electrons from an electron gun, focused and deflected by magnetic fields, hits a phosphor-coated screen which glows; used in TVs and monitors. Health hazards: X-rays (largely absorbed by the screen), UV/IR (insignificant), radio-frequency and ultrasound emissions, electrostatic field (can cause rashes), electromagnetic fields (concerns about cataracts and reproductive disorders).

LCD: smaller, lighter, no radiation problems. Top plate transparent and polarised, bottom plate reflecting; voltage applied to the crystal changes polarisation and hence colour; light is reflected, not emitted, so there is less eye strain.

Health hints: do not sit too close, avoid very small fonts, take breaks, avoid placing the screen in front of a bright window, work in well-lit surroundings; also consider posture, ergonomics, and stress.

المرجع: Lecture 03 — Slide 41-44

6. Define resolution, aspect ratio, colour depth, and anti-aliasing for bitmap displays.

عرّف resolution و aspect ratio و colour depth و anti-aliasing.

Model answer:

Resolution: used for the number of pixels (e.g. SVGA 1024 x 768) or pixel density in dpi (typically 72 to 96 dpi).

Aspect ratio: ratio between width and height (4:3 for most screens, 16:9 for wide-screen TV).

Colour depth: how many different colours each pixel can have (black/white or greys, 256 from a palette, 8 bits each for red/green/blue = millions of colours).

Anti-aliasing: softens jagged edges (jaggies, caused by the horizontal raster scan) by using shades of the line colour; also used for text.

المرجع: Lecture 03 — Slide 38-40

7. Explain large displays, situated displays (with the Hermes example), and digital paper.

اشرح الشاشات الكبيرة والشاشات الموضوعة مع مثال Hermes وورق الـ digital paper.

Model answer:

Large displays: used for meetings and lectures; technologies include plasma, video walls, projected (hand/body may obscure the screen) and back-projected (frosted glass and a projector behind).

Situated displays: displays in public places, display-only or interactive (stylus, touch); location matters because the meaning of information or interaction relates to the location. Hermes: small displays beside office doors where handwritten notes are left using a stylus and the office owner reads them using a web interface.

Digital paper: thin flexible sheets updated electronically while retaining the display, made with small spheres turned on channels with coloured liquid and contrasting spheres; a rapidly developing area.

المرجع: Lecture 03 — Slide 46-49

8. Explain virtual reality interaction: 3D positioning devices, VR headsets, and the causes of VR motion sickness.

اشرح التفاعل في الواقع الافتراضي وأسباب دوار الحركة.

Model answer:

Positioning in 3D: cockpit and virtual controls, 3D mouse (six degrees of movement: x, y, z plus roll, pitch, yaw), data glove (fibre optics detect finger position), VR helmets (detect head motion and possibly eye gaze), whole-body tracking (accelerometers or reflective dots with video processing).

3D displays: desktop VR with perspective and motion; stereoscopic vision via VR helmets or shuttered glasses; VR headsets use a small screen for each eye at slightly different angles. Simulators and VR caves project scenes on walls with real controls.

Motion sickness: time delay between head movement and display movement, and depth perception conflict (stereo distance versus all being focused in the same plane); conflicting cues cause sickness.

المرجع: Lecture 03 — Slide 50-56

9. Describe physical controls and sensing: dedicated displays, sounds, haptic devices, and environment and bio-sensing.

اوصف التحكم المادي والاستشعار: الشاشات المخصصة والأصوات والأجهزة اللمسية والاستشعار البيئي والحيوي.

Model answer:

Dedicated displays: analogue (dials, gauges, lights), digital (small LCD screens, LEDs), head-up displays (aircraft cockpits, showing the most important controls depending on context).

Sounds: beeps, bongs, clonks, whistles and whirrs, used for error indications and confirmation of actions (e.g. keyclick).

Touch, feel, smell: touch is important in games (vibration, force feedback) and simulation (feel of surgical instruments), called haptic devices; texture, smell and taste technology is very limited. Example: BMW iDrive gives small bumps for each menu item.

Environment and bio-sensing: car courtesy light switch, ultrasound detectors, RFID tags, temperature, weight, location; and body sensing such as iris scanners, body temperature, heart rate, galvanic skin response, blink rate.

المرجع: Lecture 03 — Slide 57-63

10. Explain printing, fonts, and scanning: printer types, font properties, readability, page description languages, scanners, and OCR.

اشرح الطباعة والخطوط والمسح الضوئي.

Model answer:

Printing: images made from small dots; critical features are resolution (dpi), speed (pages per minute), and cost. Dot-matrix (inked ribbon and pins, 80-120 dpi), ink-jet/bubble-jet (300 dpi or better), laser (electrostatic drum and toner, 600 dpi or better), thermal (heat-sensitive paper, simple and low maintenance).

Fonts: size measured in points (1 pt about 1/72 inch); fixed-pitch vs variable-pitched; serif vs sans-serif. Readability: lowercase makes word shape easy to read; UPPERCASE is better for individual letters and non-words such as flight numbers; serif helps on long printed lines, but sans-serif is often better on screen.

Page description languages (e.g. PostScript) send a description of the page instead of a huge bitmap. WYSIWYG aims for screen and print to match but they differ (72 dpi landscape vs 600+ dpi portrait).

Scanners convert paper to a bitmap (flat-bed or hand-held, 600-2400 dpi); OCR converts the bitmap back into text, facing issues with different fonts and page formats.

المرجع: Lecture 03 — Slide 65-75

11. Compare short-term memory (RAM) and long-term storage (disks), and explain virtual memory and compression.

قارن بين RAM والأقراص وشرح virtual memory والضغط.

Model answer:

RAM: silicon chips, ~100 ns access, usually volatile, ~100 Mbytes/s transfer; typical desktop 64 to 256 Mbytes; some non-volatile RAM stores set-up information.

Disks: magnetic (floppy ~1.4 Mbytes; hard disks 40 Gbytes to hundreds of Gbytes, ~10 ms access) and optical (lasers, more robust than magnetic; CD-ROM, DVD). Flash memory is silicon-based but persistent.

Virtual memory: stores some programs temporarily on disk to make RAM appear bigger, but swapping from disk to RAM slows things down.

Compression: lossless (exact recovery, e.g. GIF, ZIP, by finding commonalities such as 10A5B8C) and lossy (something like the original, e.g. JPEG, MP3, exploiting perception).

المرجع: Lecture 03 — Slide 77-83

12. Explain the effect of finite processing speed, Moore's law, and the myth of the infinitely fast machine. What limits interactive performance?

اشرح تأثير محدودية سرعة المعالجة وقانون مور وخرافة الآلة السريعة بلا حدود وما الذي يحد من الأداء التفاعلي.

Model answer:

Finite speed: designers assume fast processors and make interfaces more complicated, but processing cannot keep up (cursor overshooting from buffered keypresses; icon wars). Too fast can also be a problem (help screens scroll too quickly).

Moore's law: processor speed doubles every 18 months (1987: 1.5 MHz, 2002: 1.5 GHz); memory doubles every 12 months.

Myth of the infinitely fast machine: assuming no delays; good design must account for real machines (e.g. the telephone tones are an accident of implementation that was emulated in design).

Limitations: computation bound, storage channel bound, graphics bound, network capacity. Networked computing gives access to large memory and processing, other people, shared resources, but brings network delays, conflicts when many people update data, and unpredictability.

المرجع: Lecture 03 — Slide 87-92

13. Describe the history and common protocols of the Internet as presented in the lecture.

اوصف تاريخ الإنترنت والبروتوكولات المشتركة.

Model answer:

1969: ARPANET (US Department of Defense) with 4 sites; 1971: 23; 1984: 1000; 1989: 10000.

Common language (protocols): TCP (Transmission Control Protocol) is lower level, sending packets (like letters) between machines; IP (Internet Protocol) provides a reliable channel (like a phone call) between programs on machines; email and HTTP build on top of these.

المرجع: Lecture 03 — Slide 93

14. A page 11 x 8 inches is scanned at 1200 dpi with 8-bit greyscale (1 byte per pixel), uncompressed. Estimate the file size. [Calculation]

صفحة 11 × 8 بوصة اتمسحت بدقة 1200 dpi ويتدرج رمادي 8 بت. قُدِّر حجم الملف بدون ضغط.

Model answer:

Pixels along height = $11 \times 1200 = 13,200$; along width = $8 \times 1200 = 9,600$.

Total pixels = $13,200 \times 9,600 = 126,720,000$ pixels.

At 1 byte per pixel = about 126.7 Mbytes. The lecture quotes roughly 128 Mbytes for this example (rounded).

المرجع: Lecture 03 — Slide 81

15. Calculate the data rate of uncompressed video with resolution 512 x 512, 12-bit colour, and 25 frames per second. [Calculation]

احسب معدل بيانات فيديو غير مضغوط بدقة 512×512 ولون 12 بت و25 إطار/ثانية.

Model answer:

Pixels per frame = $512 \times 512 = 262,144$.

Bits per frame = $262,144 \times 12 = 3,145,728$ bits = 393,216 bytes.

Per second = $393,216 \times 25 = 9,830,400$ bytes/s, about 10 Mbytes per second (matches the lecture).

المرجع: Lecture 03 — Slide 81

16. Using Moore's law (processor speed doubles every 18 months), check the lecture's claim that a PC went from 1.5 MHz in 1987 to about 1.5 GHz in 2002.

[Calculation]

باستخدام قانون مور تحقق من أن سرعة الحاسوب وصلت من 1.5 MHz عام 1987 إلى حوالي 1.5 GHz عام 2002.

Model answer:

Time span = 2002 - 1987 = 15 years = 180 months.

Number of doublings = 180 / 18 = 10.

Growth factor = $2^{10} = 1024$; 1.5 MHz x 1024 = 1,536 MHz, about 1.5 GHz. This matches the lecture.

المرجع: Lecture 03 — Slide 89

17. Using the run-length idea from the lecture, compress the string AAAAAAAAAAABBBBBCCCCCCCC. How many characters are saved, and is this lossless or lossy? [Calculation]

باستخدام فكرة الضغط في المحاضرة اضغط السلسلة AAAAAAAAAAABBBBBCCCCCCCC وكام حرف اتوفر وهل الضغط lossless ولا lossy؟

Model answer:

Count runs: A x 10, B x 5, C x 8.

Compressed form: 10A5B8C (6 characters) versus the original 10 + 5 + 8 = 23 characters, saving 17 characters.

It is lossless compression, because the exact original can be recovered by expanding the counts (like GIF and ZIP).

المرجع: Lecture 03 — Slide 83